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Ans 1. Newton's second law: rate of change of momentum is proportional to applied force, in direction of force. $F = ma$.

Ans 2. Speed is scalar (magnitude only). Velocity is vector (magnitude + direction). Example: 40 km/hr north.

Ans 4. SI unit of current is Ampere (A). One ampere = one coulomb of charge per second.

Ans 6. $u=0$, $a=2$, $t=5$. $v=u+at=10$ m/s. $s=ut+\frac{1}{2}at^2=25$ m.

Ans 8. Ohm's law $V=IR$. $V = 2 \times 10 = 20$ Volts.

Ans 11 (a). $v=u+at$... average velocity $= (u+v)/2$, $s = \text{avg} \times t$, substituting:
 $s = ut + \frac{1}{2}at^2$

Ans 9. Three states: solid, liquid, gas. Particle spacing/arrangement differs between each.

Ans 7. Mitosis: 2 identical daughter cells (growth/repair). Meiosis: 4 cells, half chromosomes (reproduction).

Ans 12. Rainwater harvesting; fixing leaking pipes/taps promptly to avoid wastage.

formula sheet (pencil, faint): $F=ma$, $v=u+at$, $s=ut+\frac{1}{2}at^2$, $V=IR$, $P=VI$